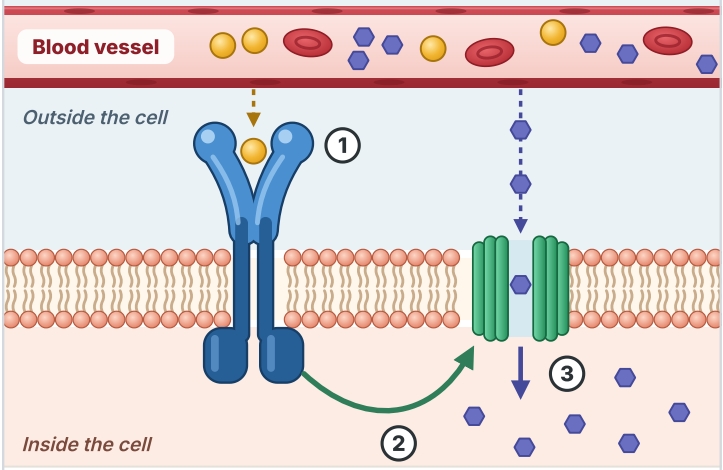
Patient name _____ Date _____

Understanding Insulin Resistance

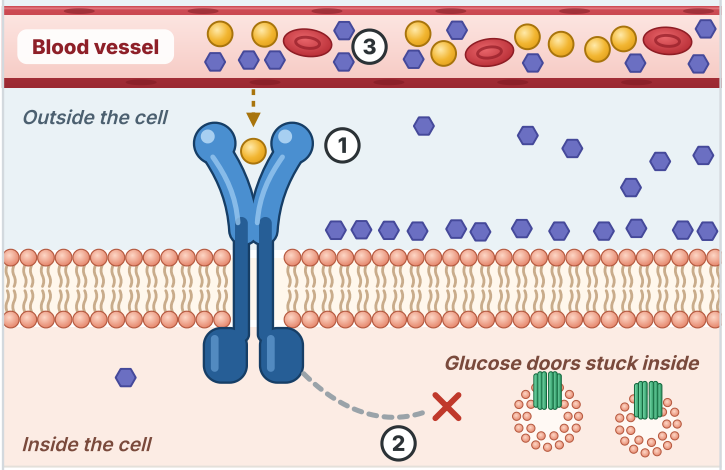
Insulin is a hormone your pancreas releases, especially after you eat. It is a messenger: it tells your cells to open their **glucose doors**, so sugar (glucose) can leave your blood and go into the cells for fuel. With **insulin resistance**, the cells stop listening well. Sugar backs up in the blood, so the pancreas makes more and more insulin to keep up. This can go on quietly for years before blood sugar rises high enough to be called prediabetes or type 2 diabetes.

NORMAL — *cells answer insulin*



1. Insulin from the blood fits into its receptor.
2. The signal opens the glucose door.
3. Glucose moves into the cell for fuel.

INSULIN RESISTANCE — *cells barely answer*



1. Insulin still fits its receptor, but the cell barely listens.
2. The signal is weak, so the glucose doors stay inside.
3. Glucose stays in the blood, so blood sugar rises.

KEY Glucose (sugar) · Insulin · Insulin receptor · GLUT4 glucose door · Cell membrane · Red blood cell

WHAT DRIVES IT

- Sugary drinks and processed food
- Sitting most of the day
- Short sleep, long-term stress
- Extra fat around the belly
- Family history, age 45 or older
- Some medicines, such as steroids

CLUES TO WATCH FOR

- Often no signs at all
- Weight gain at the waist
- Tired or foggy after meals
- Cravings, hungry soon after eating
- Dark velvety skin patches, skin tags
- High triglycerides, low HDL

WHERE IT CAN LEAD

- Prediabetes, then type 2 diabetes
- Heart disease and stroke
- High blood pressure
- Fatty liver (MASLD)
- PCOS and hormone shifts
- Memory decline

MORE THAN 1 in 3 U.S. adults have prediabetes, and **more than 8 in 10** of them do not know it (CDC). Insulin often climbs years before blood sugar does, so ask your pharmacist about a **fasting insulin** test along with the numbers below.

Know your numbers

Write your results in the boxes, then go over them with your pharmacist.

Fasting blood sugar glucose, mg/dL, after 8 hours with no food

NORMAL	PREDIABETES	DIABETES	My result
under 100	100 to 125	126 or more	

A1c average blood sugar over about 3 months

NORMAL	PREDIABETES	DIABETES	My result
under 5.7%	5.7 to 6.4%	6.5% or more	

Metabolic syndrome check Tick each sign you have. **Any 3 of these 5** means metabolic syndrome. Taking medicine for one counts as a yes.

- ☐ **Waist** — 40 in or more (men); 35 in or more (women)
- ☐ **Triglycerides** — 150 mg/dL or more
- ☐ **HDL (good) cholesterol** — under 40 (men); under 50 (women)
- ☐ **Blood pressure** — 130/85 or higher (either number)
- ☐ **Fasting blood sugar** — 100 mg/dL or more

Sources: American Diabetes Association (a diabetes result is confirmed with a repeat test); NHLBI and American Heart Association.

Lifestyle-Based Medicine

Low-glycemic Mediterranean plate

- Protein first at every meal
- Half the plate vegetables
- Whole-food carbs, paired with protein or fat
- Three to five balanced meals
- Limit sugar and sweet drinks

Lower stress, better sleep

- Commit to a simple plan
- Deep breathing, 5 minutes a day
- 7 to 9 hours of sleep, same times daily
- One 10-minute mindful walk a day

Move your muscles

- Walk 10 minutes after meals
- Strength training 2 to 3 times a week
- Stand up every 30 to 60 minutes
- Moderate to hard exercise 3 to 5 days a week

Nutrient Solutions

What your body needs, and the NutriDyn formula that delivers it

All-in-one glucose support

dihydroberberine, chromium, ALA, cinnamon

START HERE



Gluco IR

150 capsules · 30 servings

- Supports healthy glucose metabolism
- Supports insulin sensitivity

IN EACH SERVING

100 mg dihydroberberine (GlucoVantage), 300 mg alpha lipoic acid, 500 mg Ceylon cinnamon, 600 mcg chromium, 200 mg gymnema, green tea

5 capsules daily
with meals

Metabolic signaling

berberine



Berberine Pro

90 capsules

- Supports healthy glucose levels already in a normal range
- Supports healthy lipid levels

IN EACH SERVING

100 mg dihydroberberine + 84 mg organic cinnamon extract per capsule

1 capsule 3 times daily
before meals

Insulin's helper mineral

magnesium



Magnesium Glycinate

60, 120, 160 or 320 capsules

- Needed for normal insulin signaling
- Low magnesium is common with high blood sugar

IN EACH SERVING

260 mg magnesium per 2 capsules, Albion chelated (di-magnesium malate, lysinate glycinate)

2 capsules, 1 to 2 times daily
with food

Balance fats

omega-3 fatty acids



Omega Pure EPA-DHA 1000

120 softgels

- Supports healthy triglyceride levels
- Supports a healthy inflammatory response

IN EACH SERVING

1,000 mg EPA + DHA per softgel (730 mg EPA, 270 mg DHA)

1 softgel daily
with your evening meal

YOUR DAILY RHYTHM

Morning
Gluco IR 3 capsules
Berberine Pro 1 capsule

Evening meal
Gluco IR 2 capsules
Berberine Pro 2 capsules
Omega Pure EPA-DHA 1000 1 softgel

Bedtime
Magnesium Glycinate 2 capsules

The full plan is **11 pills a day**. Overwhelmed? Start with the **Start here** pick, then add one product a week.

PHARMACIST CHECK

- **Metformin** lowers B12 over time. Ask for a B12 check every year.
- **On diabetes medicine?** Berberine adds to sugar-lowering effects: check your glucose and tell your prescriber.
- **Berberine interacts** with some medicines (for example cyclosporine). Let your pharmacist review your list.

Special considerations for you:

Recommended by _____ Phone _____

◆ These statements have not been evaluated by the Food and Drug Administration. These products are not intended to diagnose, treat, cure, or prevent any disease. Nutrient support works alongside, never instead of, the care and medicines your practitioner prescribes.

NutriDyn